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Experiment 7: ARM C-Interfacing - Interfacing of DIP Switch, LED, Stepper Motor and their Control

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1 Aim

Using C-interfacing, use C-programming, to implement the following tasks:

(i) Read the status (binary position) of the DIP (Dual Inline Package) switch and use the LEDs (8 LEDs are provided) to display the state (ON or OFF) of each of the 8 DIP switches.

(ii) Stepper motor control using LPC2148 based ARM development board. Due to ongoing pandemic, only the emulated version of this experiment is intended here.

2 Equipments, Hardwares / Softwares Required

The list of equipments, components required are:

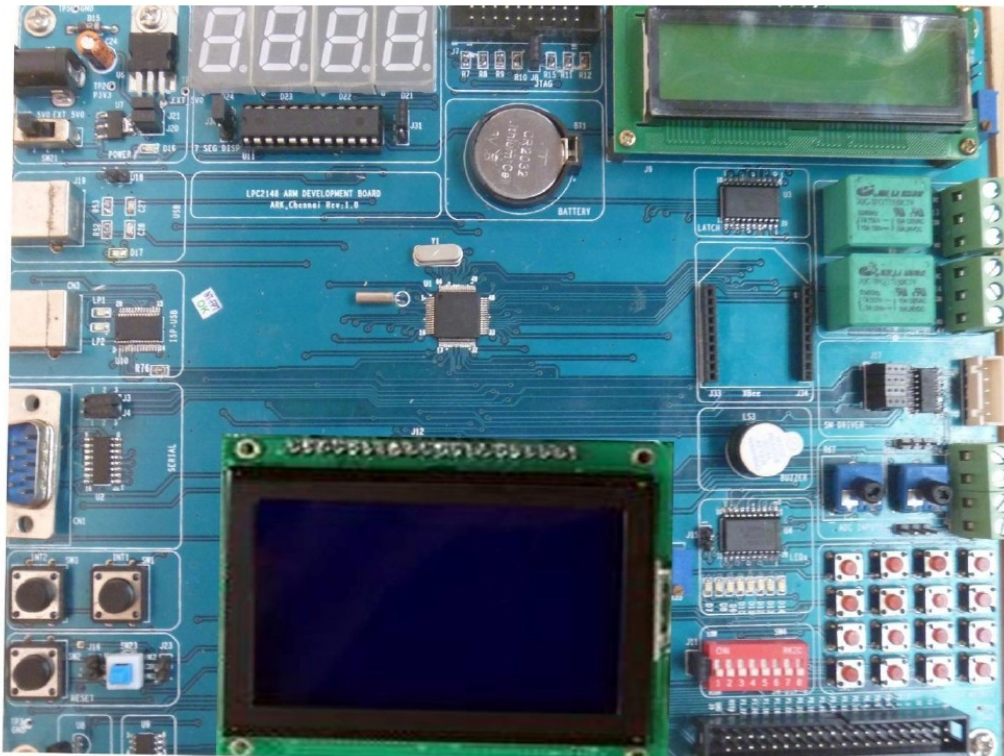
1. ARK's LPC2148 embedded ARM development board and accessories
2. RS-232 cable
3. Keil microvision 5
4. USB - serial converter (this is a must when the PC loaded with keil doesnt have a serial port).
5. flash magic
6. Burn o-mat
7. Stepper motor

3 Background Information

In this Section, we would discuss the following background information, which are very much essential to do the above experiment

1. ARK's ARM based LPC 2148 Development Board
2. DIP & LED Interfacing in LPC 2148 ARM processor
3. Stepper motor control by LPC 2148 ARM processor
4. Demo programs (which could be used to understand and write the C code for the tasks defined at the end).

3.1 ARK's ARM7 based LPC2148 Development Board - Anatomy



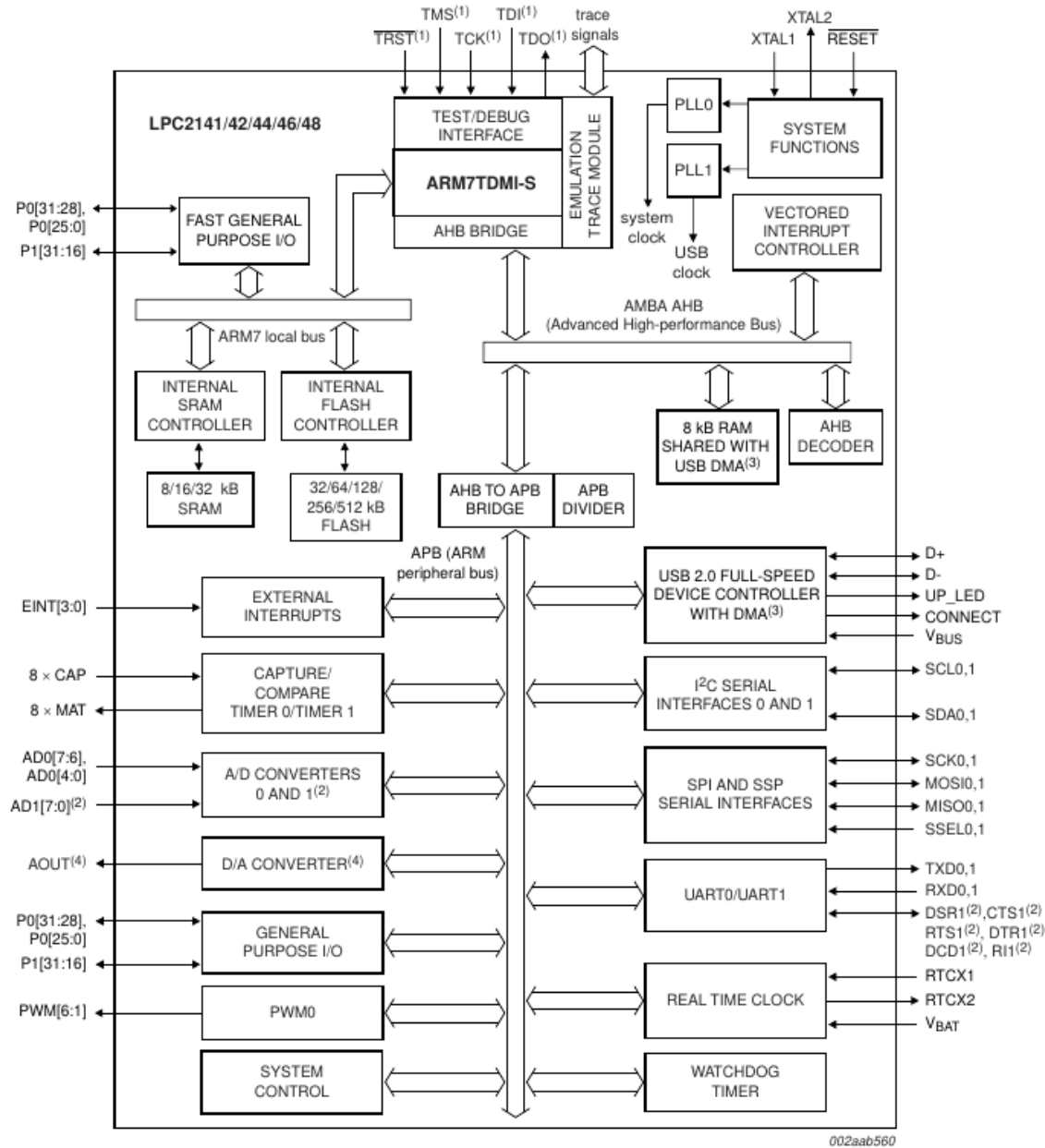
The microprocessor used in the microcontroller LPC2148 development board, ARM7TDMI-S core or CPU, whose expansion of the terminology ARM7TDMI-S is given below. LPC2148 is from NXP semiconductors while LPC stands for Low Power Consumption.

3.1.1 The ARM and ARM7TDMI-S

The ARM CPU core used inside the LPC2148 is called the ARM7TDMI-S (where T stands for Thumb Instructions, D for on-chip debugging support, M for multiplier, I for embedded In-circuit emulation hardware and S for the synthesizable option based on RTL provided). ARM7TDMI was the first of a range of processors introduced in 1995 by ARM Holdings, UK. ARM7TDMI is the first core to include the Thumb instruction set. The Thumb instruction set contains 16-bit instructions. These serve as a 'compact shorthand' for a subset of the 32-bit instructions of the ARM. For 32-bit systems, ARM instructions are recommended for higher performance. Thumb instructions are appropriate for memory constrained systems.

3.1.2 Block Diagram of LPC2148

The LPC2148 block diagram is shown in Fig. below.



The instructions for the board to be used in the experiments are available as a separate pdf file.

3.1.3 Brief Description of the Software Environment

In this laboratory experiment, programming will be done in C. The instructions for writing C language programs in Keil μ Vision are as follows.

1. Open Keil μ Vision and Click project > New μ Vision Project
2. Select the 'Legacy device Database' under device dropdown. Search for LPC2148 and select it
3. Choose Yes when prompted to copy LPC2100.s Startup file
4. Right Click on Target 1, choose 'Options for Target 1'.
5. Under the Target tab tick 'Use Microlib'. Under Output tab tick 'Create Hex File'
6. Right click Source Group1 under Target in the left side of the keil window. Select Add new Item to group.
7. Select C file in the list, give a filename and save.

8. Write your program.
9. Go to Project tab and click on Build Target (or press F7).

3.1.4 How to Upload the Program to the Flash on LPC2148 in the Board

The instructions for uploading the program on to the flash memory on the LPC2148 are as follows.

1. Run FlashMagic.
2. Select LPC2148.
3. Choose the correct COM port to which the USB to Serial converter is connected. If you do not know the COM Port, use Device Manager to find out.
4. Set baud rate to 19200 and Oscillator to 12 MHz.
5. Choose the Hex file to be transferred.
6. On the board, put the "Program/Execute" slide Switch to "Program" mode (slide it down)
7. On FlashMagic click 'Start' to start programming.
8. Once programming is done, put the "Program/Execute" slide Switch to "Execute" mode (slide it up) and press reset to execute the program.

3.2 LEDs and DIP switches in LPC-2148 Board

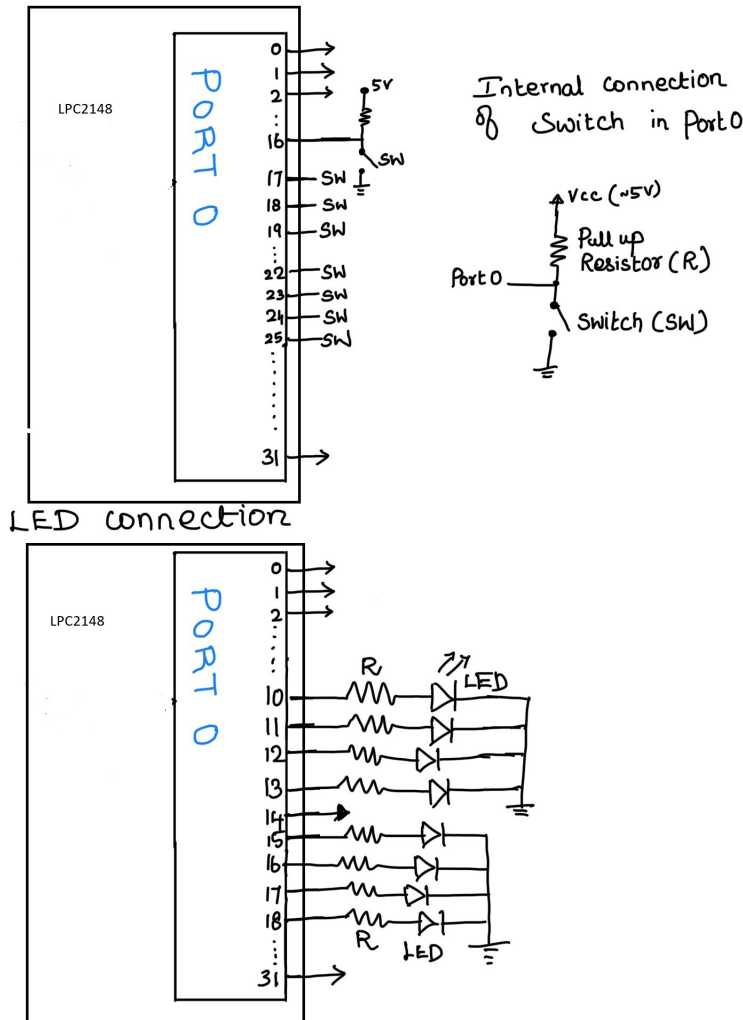
LEDs and DIP switches are most common in development boards. In , DIP switches are the red ones in bottom right (just before the key pad button matrix). And LEDs are just above the DIP switches.

3.2.1 LEDs

Light Emitting Diodes are the most commonly used components usually for displaying Pin's digital state. LPC-2148 has 8 LED's that are connected to the microcontroller port line. One has to close the jumper J15 to use LEDs (See ARK Embedded System Lab Manual for LPC2148).

Used port lines: Refer the following connection diagram to connect DIP switches and LEDs to the LPC2148 [In lab, the student doesn't have to do the connections, as these are already hardwired in ARK's ARM based LPC2148 development board].

LPC2148 Switch & LED connection in kit



Jumper J11 is to be used for enabling DIP switches. DIP switches SW0 - SW7 are connected to PORT1 with LPC2148 hardware pins: P1.17 to P1.25 (with P1.20 excluded) and LEDs LED0 - LED7 are connected to PORT0 with LPC2148 hardware pins: P0.10 - P0.18 (with P0.14 excluded).

Open and run, "TStLED-InLPC2148Mdle.c", the c code uploaded in the moodle. Note that the LEDs are 'hard-wired', to PORT0 of LPC2148.

3.2.2 Switches

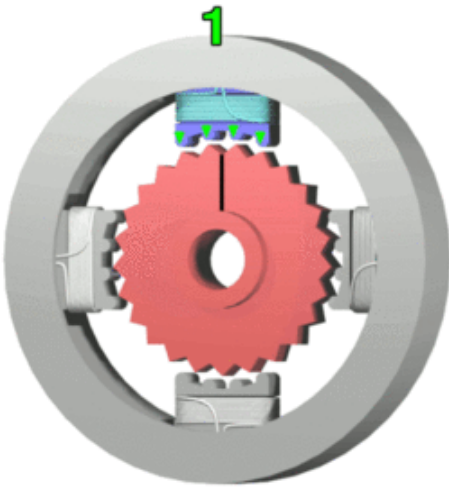
Switches are devices that have two positions - ON and OFF, which have a toggle to establish or break a connection between two contacts. The LPC-2148 development board has 8 one-way (ON - OFF) DIP switches.

Used port lines: Refer the above connection diagram for connecting DIP switches to LPC2148 hardware pins.

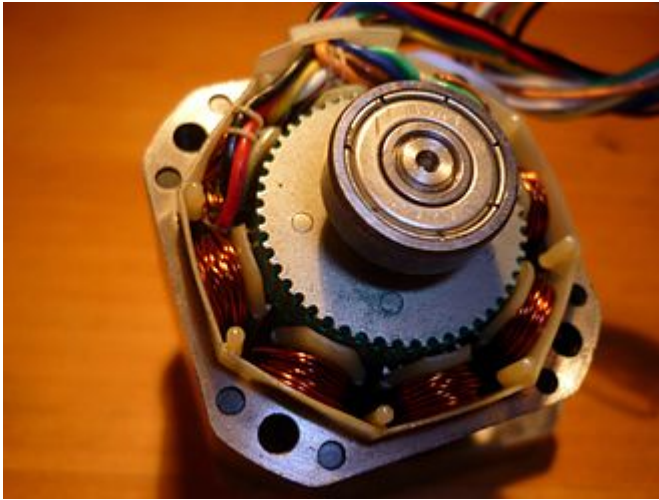
3.3 Stepper Motor

A stepper motor simply rotates by a fixed angle when the voltage is applied.

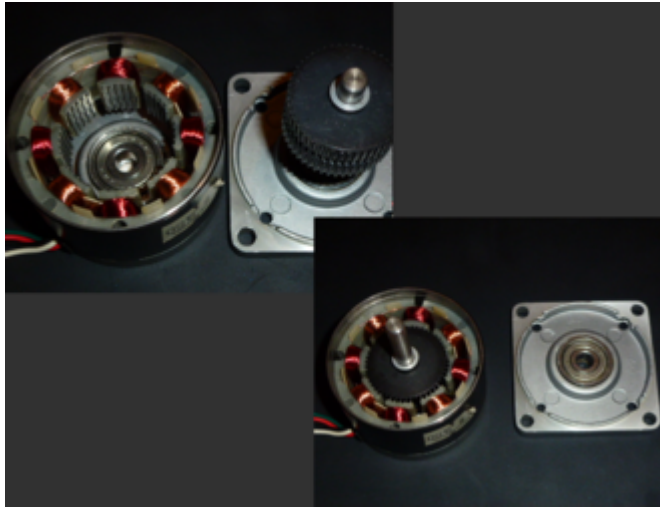
A stepper motor is a brushless electric DC motor with a large number of poles (on both rotor and stator), such that a full rotation is divided into a number of equal steps. The motor's position can then be commanded to move and hold at one of these steps without any position sensor for feedback (on an open-loop controller), as long as the motor is carefully sized to the application in respect to torque and speed.



Stepper motors effectively have multiple "toothed" electromagnets arranged around a central gear-shaped piece of iron. The electromagnets are energized by an external driver circuit or a micro controller. To make the motor shaft turn, first, one electromagnet is given power, which magnetically attracts the gear's teeth. When the gear's teeth are aligned to the first electromagnet, they are slightly offset from the next electromagnet. This means that when the next electromagnet is turned on and the first is turned off, the gear rotates slightly to align with the next one. From there the process is repeated. Each of those rotations is called a "step", with an integer number of steps making a full rotation. In that way, the motor can be turned by a precise angle.



The circular arrangement of electromagnets is divided into groups, each group called a phase, and there is an equal number of electromagnets per group. The number of groups is chosen by the designer of the stepper motor. The electromagnets of each group are interleaved with the electromagnets of other groups to form a uniform pattern of arrangement. For example, if the stepper motor has two groups identified as A or B, and ten electromagnets in total, then the grouping pattern would be ABABABABAB.



Electromagnets within the same group are all energized together. Because of this, stepper motors with more phases typically have more wires (or leads) to control the motor.

3.3.1 Used Port lines

The 4 wires from stepper motor (colors, blue, pink, yellow, orange) to be connected to P0.4 to P0.7 of LPC2148. The last red wire is to be connected to motor Vcc. The universal control sequence to be given to the input of the above four wires are {0x09, 0x0C, 0x06, 0x03}.

3.3.2 Stepper Motor Driver in ViARM-2378

3.3.3 Jumper Position

Closed 1 & 2 - Internal voltage for stepper motor

Closed 2 and 3 - External voltage for stepper motor

3.4 Demo Programs

These programs are presented here to enable the students write their own program independently, given an engineering problem of similar type.

3.4.1 DIP Switch & LEDs

The c code, for reading the DIP switch state and displaying it in the LEDs are implemented in the demo code, DIP-SwchToLED-InLPC2148.c uploaded in moodle. The above code assumes:

1. The DIP switches are connected to PORT1 with LPC2148 pins as above. The LED switches are connected to PORT0 with LPC2148 pins as above.
2. Note that the DIP switch (red colored stuff), when the position is 'UP', towards the LPC2148 chip then it is ON (meaning closed). This means that the pull-up resistor is grounded and hence output is ground (0 V). When it is down (pointing towards the periphery), then output is Vcc (logical '1'). Accordingly the sw_status (32-bit unsigned constant (integer, binary)) carries the state of all DIP switches at any time slot, 't'. Hence, when the DIP SW is ON (UP, closed), LEDs should be OFF, which is the convention followed in the above code.

3.4.2 Stepper Motor Control

The demo program (StpprMtrCntrlInLPC2148.c) is uploaded in the moodle, controls the stepper motor (angle to which it has to be turned etc). The universal control sequence to control the stepper motor is given above and is implemented in the above code. The delay between each step depends on the frequency of the microcontroller, LPC2148. Very common operating frequency of LPC2148 is 60 MHz and accordingly the code could be changed to realize 1 ms per single run of (while()) loop.

4 Problem Definitions

Following are the tasks you need to take up for this lab session:

4.1 Tasks Involving LED

4.1.1 Task 1: Only LED

Write a C-program to display last three digits of your roll number (in hex) in LEDs on the ARM-board. (Your roll number is in decimal, note).

4.1.2 Task 2: DIP Switch and LED

Read the DIP positions and display the byte in LED (logical '1' by OFF & logical '0' for 'ON'). You can see the ON position written on the red colored DIP switches.

4.1.3 Task 3: Sum

Write a C program to read a DIP switch, split into two nibbles (4 bits), display the sum in the LEDs.

4.1.4 Task 4: Product

Write a C program to read a DIP switch, split into two nibbles (4 bits), display the product in the LEDs.

4.2 Tasks Involving Stepper Motor

1. Modify the demo code (StpprMtrCntrl.c) supplied to demonstrate the control of stepper motor to rotate in opposite direction.
2. Gauge the speed of the stepper motor
3. What is the max speed with which the stepper motor can rotate?
4. How many steps are required for 360 deg rotation

5 Procedure

Stepper Control

1. Write a C program, which could control the stepper motor
2. Edit the above program file in Keil software. In Keil software one can edit, recompile and run etc. Compile it in Keil platform
3. Connect the serial cable from ARM LPC2148 and PC
4. Dump it in ARM LPC2148
5. Connect the stepper motor [Lab staffs / TAs would have done all these including jumper wire settings, so that you need not worry about these].
6. Run the program on the development kit by resetting the ARK LPC2148.

For emulation you only need to do the following:

1. Write a C program, which could control the stepper motor
2. Edit the above program file in Keil software. In Keil software one can edit, recompile and run etc. Compile it in Keil platform
3. Identify the (digital) signal and demonstrate it to the TA

6 Results

1. Run the program and ask the TA to see the output
2. Take a snapshot using your mobile, write a report and submit